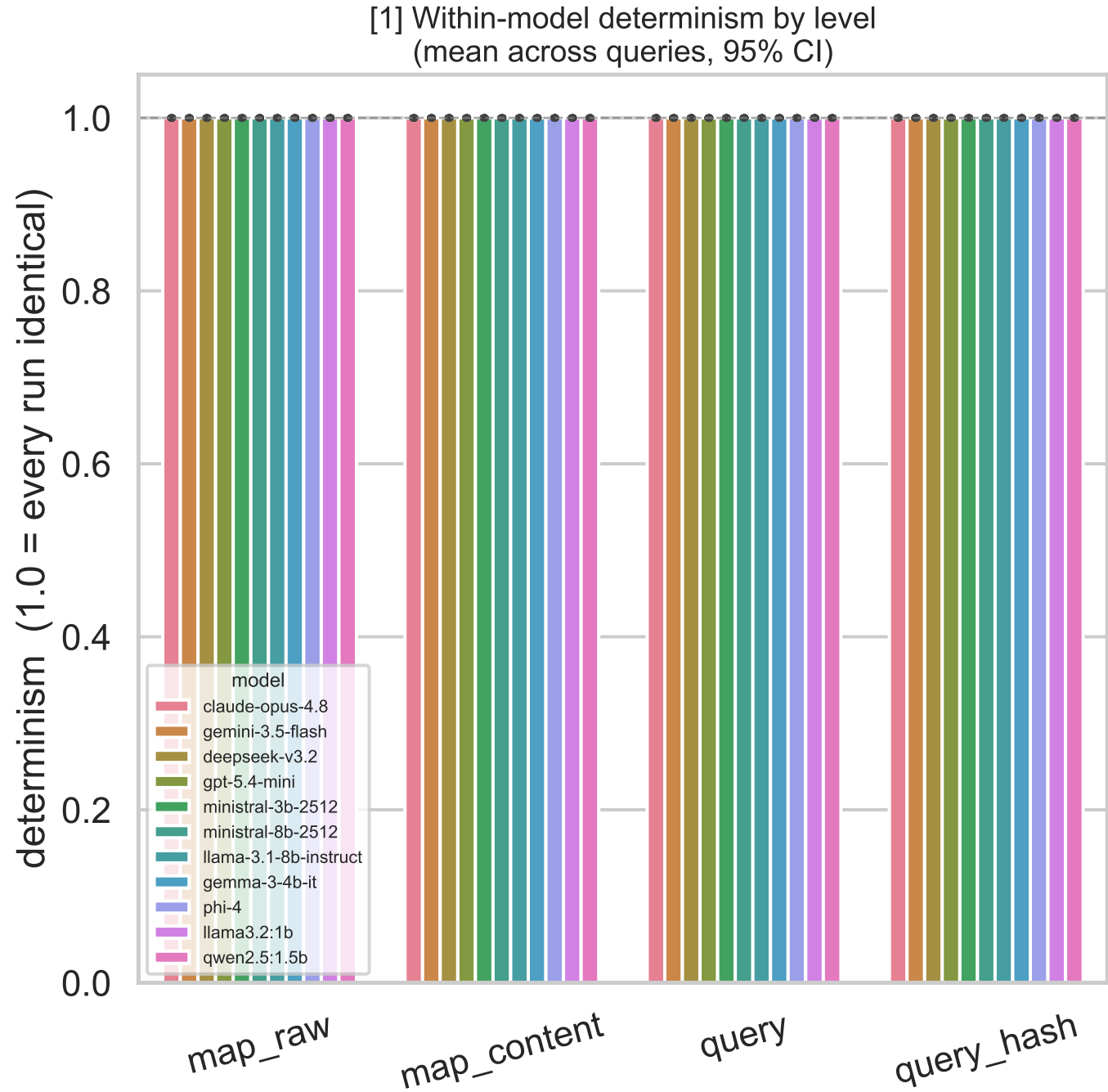
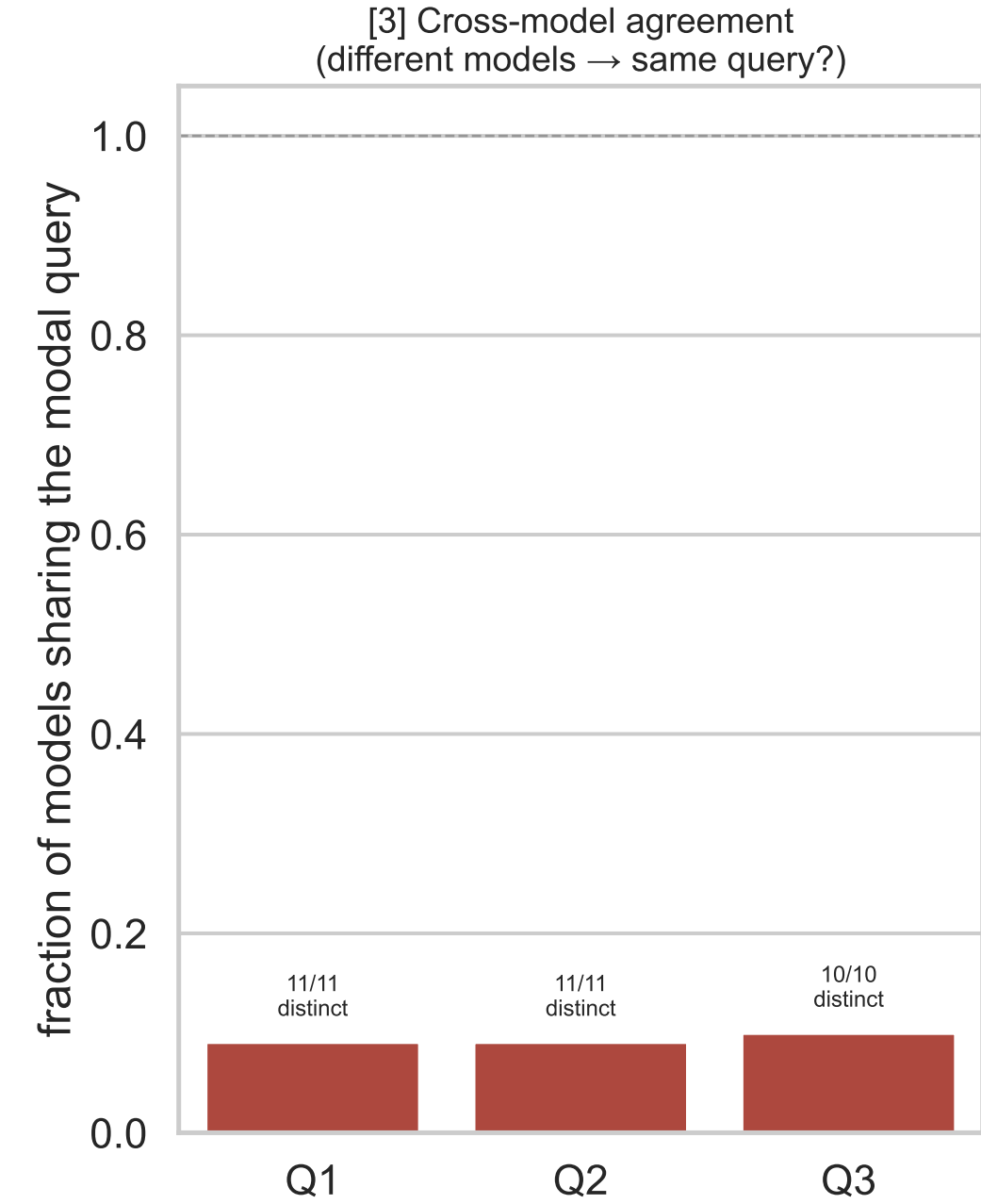




[2] query_hash reproducibility
(same model, 10 runs → same query?)

claude-opus-4.8	1.00	1.00	1.00
deepseek-v3.2	1.00	1.00	1.00
gemin-3.5-flash	1.00	1.00	1.00
gemma-3-4b-it	1.00	1.00	1.00
gpt-5.4-mini	1.00	1.00	1.00
llama-3.1-8b-instruct	1.00	1.00	1.00
llama3.2:1b	1.00	1.00	1.00
ministral-3b-2512	1.00	1.00	1.00
ministral-8b-2512	1.00	1.00	1.00
phi-4	1.00	1.00	1.00
qwen2.5:1.5b	1.00	1.00	1.00
	Q1	Q2	Q3

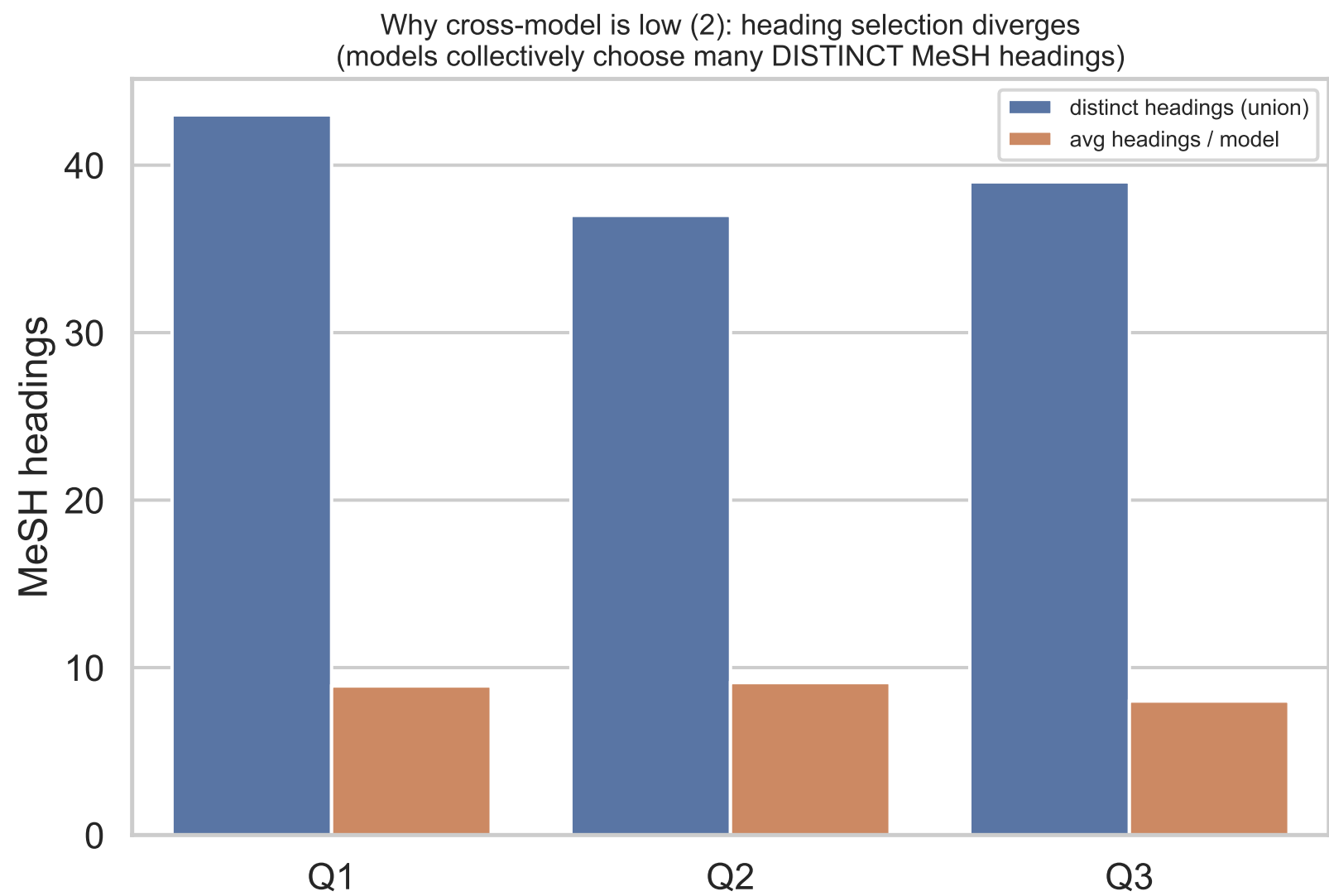
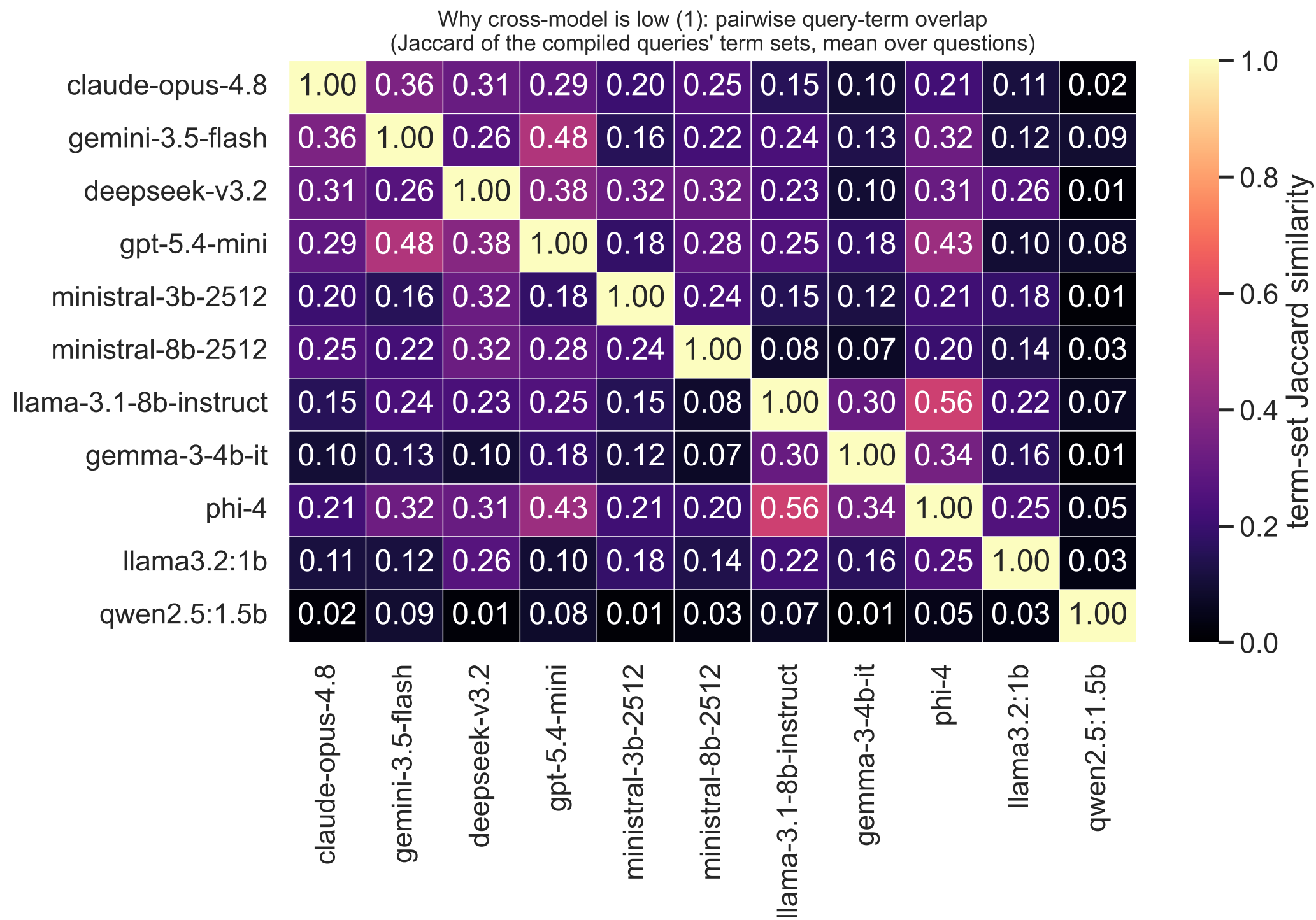


QUESTIONS: Q1 = Does optogenetic stimulation of the hippocampus... Q2 = single-cell RNA sequencing of microglia in Alzheimer's... Q3 = CRISPR-Cas9 off-target effects detection method...

LEVELS — map_raw: LLM JSON byte-identical across runs · map_content: same ignoring key/list order · query: final compiled PubMed query · query_hash: SHA-256 of that query.

ARTIFACT — query_hash is the reproducibility unit: a 16-hex digest of the exact query string. Identical hash => byte-identical query => PubMed returns the identical article set. It is what you cite in a protocol to prove a search can be reproduced. [1][2] WITHIN-model = one model repeats itself (=1.00 here → reproducible). [3] CROSS-model = different models agree (mean 0.0939; low — see page 2).

Cross-model divergence — the SAME question, different MeSH headings per model



Mean pairwise term-set overlap across models = 0.198 (0 = no shared terms, 1 = identical). Each model independently DECOMPOSES the question and proposes different MeSH headings; those resolve to different descriptors, whose exploded entry-terms differ, so the compiled query (and its SHA-256) differs. The deterministic layer faithfully reproduces whatever headings a model picked — it cannot make two different models pick the same ones. Hence reproducibility is PER-MODEL: to reproduce a search, pin the (model + question) and keep strict+cache.

Definitions & methodology

PIPELINE

research question → LLM /api/map (temperature 0; the ONLY non-deterministic step) proposes concept blocks
+ candidate MeSH headings → each heading resolved against a local MeSH index (hallucinations drop out) →
strict free-text derived from the index → query_builder compiles a Boolean PubMed query → SHA-256 hash.

QUESTIONS TESTED

- Q1: Does optogenetic stimulation of the hippocampus improve memory consolidation in rodent models?
- Q2: single-cell RNA sequencing of microglia in Alzheimer's disease
- Q3: CRISPR-Cas9 off-target effects detection methods

DETERMINISM LEVELS (score = size of the most common output group / N runs; 1.0 = all runs identical)

map_raw	the LLM's raw JSON, byte-for-byte – catches any wording/ordering change
map_content	that JSON with keys and lists sorted – catches real intent/keyword drift, ignores reordering
query	the final compiled PubMed query string
query_hash	SHA-256 (16 hex) of that query – THE ARTIFACT: identical hash => identical query => identical PubMed result set. This is what a systematic-review protocol cites to prove reproducibility.

TWO KINDS OF DETERMINISM

WITHIN-model (reproducibility)	same model, run N times → same query_hash? Here 1.00 for every model, achieved by: strict MeSH-derived terms + pinning the map (run once, reuse).
CROSS-model (agreement)	different models → same query_hash? Here ≈ 0.09 (essentially none).

WHY CROSS-MODEL IS LOW (and why that's fine)

Each model independently decomposes the question and picks its OWN set of MeSH headings. Different headings → different descriptors → different exploded entry-terms → different query → different hash. The deterministic layer reproduces faithfully whatever a model chose; it cannot force two models to agree. So reproducibility is PER-MODEL: record the model + question alongside the query_hash, and any reader can reproduce the search.